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E-303 UNIT AUXILIARY TRANSFORMERS

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of Unit Auxiliary Transformers in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODE AND STANDARDS

The Contractor shall comply with the requirement of Part 4, Subsection 2.5 Codes and Standards. The specific standards for the equipment are called out in Technical Specification E-301.

3. OPERATING AND DESIGN CONDITIONS

The transformers shall be suitable for outdoor operation at rated power in a power plant on a tropical sea coast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for detailed information on meteorological, seismic and environmental conditions at the site.

Transformers will be used in step-down service to connect a steam turbine generator to its unit auxiliary power system, respectively. Therefore, the transformers may experience periods of overvoltages and over/under frequency following a trip-out (from rated conditions) of the unit, disconnecting the transformers from the high voltage system.

The Contractor shall state, any limitations with respect to the length of time that his transformers can operate satisfactorily at overvoltage and/or over/under frequency conditions following such a trip.

3.1 Short Circuit Capability

The transformers may experience unusually severe and extended mechanical stresses during short circuits on the external low voltage.

Transformers shall be capable of withstanding, without injury, the mechanical and

thermal stresses resulting from a short circuit external to the low voltage winding when the short circuit is as follows:

- (1) Symmetrical rms short circuit MVA at instant of short circuit:
 - a. H-winding*)
 - b. X-winding*)

- (2) Peak asymmetrical short circuit current:
 - a. H-winding37 (times normal) *)
 - b. X-winding37 (times normal) *)

- (3) Symmetrical rms current for extended thermal and mechanical stresses:

	High Voltage Winding	Low Voltage Winding
Initial short circuit current.....(times normal) *)	14 *)	14*)
This initial value will continue for.....(seconds)	0.8 *)	0.8 *)
At which time the current will rise to(times normal) *)	16.7*)	26.7 *)
From which value it will decay linearly to zero in(seconds) after fault in- ception	8.8 *)	8.8 *)

NOTES : *) The normal value of current is the rms value of winding current when the winding is carrying its maximum rated MVA at the rated voltage. Value given here is minimum value. The Contractor shall give rating calculations depending on his generator characteristics to be approved by the Owner.

4. DETAILED REQUIREMENTS

4.1 General

The transformers shall conform to the requirements in Technical Specification E-301 except as modified herein.

The transformers shall be three-phase, two-winding, 50 Hz, outdoor, single tank,

oil-immersed type, cooling ONAN or ONAN/ONAF as shown in the Bid drawing document.

The low voltage winding neutral shall be fully insulated.

4.2 Transformer Ratings

	High Voltage Winding	Low Voltage Winding
- Nominal system voltage (line to line) ...(kV)	Generator Terminal Voltage	6.3
- Transformer connections (Wye or delta)	Delta	Wye
- Rated winding voltage(kV)	Generator Terminal Voltage	6.3
- Insulation class	A	A
- Transformer Rated Load:		
Self cooled, ONAN.....(MVA)	*)	*)
- Tap changing method : Off-load tap changer		
a. Above rated voltage at minimum rated MVA capacity (Number of taps and percent volts each taps)	2 x 2.5%	---
b. Below rated voltage at maximum rated MVA capacity (Number of taps and percent volts each taps)	2 x 2.5%	---
- System neutral (Isolated or type of grounding)	---	Grounded through a resistor

NOTES : *) Value shall be determined by the Contractor based on his equipment. Calculations ratings shall be submitted for approval by the Owner. The rated load shall include the auxiliary for the gas turbine plus the steam turbine and for the other such as crane, lighting, ventilation, compressor and all equipment under scope of this Contract.

	High Voltage Winding	Low Voltage Winding
System highest voltage(kV)	Generator High-	7.2

	High Voltage Winding	Low Voltage Winding
	est Voltage	
Winding basic impulse insulation level (BIL) (kV _{peak})	*)	60
Power frequency withstand voltage(kV _{rms/1 min})	*)	20

4.2.1 Tapping Requirements

A manually operated tap changer shall be provided for operation when the transformer is not excited. Tap changer hand wheel shall not be higher than one (1) meter above transformer base.

4.2.2 Impedance

The impedance at the rated connections shall be 7.5-10.5 percent (exact value shall be determined by the Contractor to meet short circuit and voltage regulation requirement and to be submitted to the Owner for approval) with standard tolerance at 75°C.

4.2.3 Angular Displacement

The transformer shall have a standard phase relationship, vector group DYn 11.

4.3 Terminal Connection and Bushings

Transformer leads shall be brought out of the transformer case by means of bushings of the outdoor type. Bushing shall be capable of withstanding the 50 Hz applied voltage tests and the lightning and switching impulse tests specified.

4.3.1 Bushing Ratings

	High Voltage Winding	Low Voltage Winding & Neutral
System highest voltage(kV)	Generator Voltage	7.2
Basic lightning impulse insulation level (BIL)(kV _{peak})	*)	60
Creep distance(mm)	*)	276
Wet power frequency withstand voltage	*)	20

	High Voltage Winding	Low Voltage Winding & Neutral
.....(kV _{rms.})		

NOTES : *) The value shall be in accordance with IEC 60071 and 60076.

The color of the bushings shall be chocolate brown, wet process porcelain.

Bushings for the high-voltage windings shall be top mounted and have a flanged enclosure and adapters suitable for connection to isolated phase bus duct. The low point of flanged enclosures shall be provided with a 15 mm diameter screened drains.

Bushings for the low voltage windings shall be top or sidewall mounted and have flanges suitable for connection to a non-segregated phase bus duct. The Contractor shall furnish a low voltage bushing terminal adapter.

The low voltage neutral bushing shall be cover mounted with terminal adapter.

4.4 Bushing Current Transformers

The Contractor shall design and provide bushing current transformers for metering and protection relaying in accordance with the Technical Specifications E-250

Exact ratings shall be determined by the Contractor based on his equipment requirements. Rating calculation shall be submitted to the Owner for approval.

4.5 Neutral Grounding Resistor

A neutral grounding resistor shall be furnished for the low-voltage winding on each transformer. The resistor shall be arranged for cover mounting adjacent to the low voltage neutral bushing.

The resistor shall be a cast iron grid, enclosed-tank type and shall be corrosion protected. Steel mounting pad shall be hot dipped galvanized.

The resistor shall conform to ANSI/IEEE 32 or IEC Standard requirements, terminology, and test procedure for neutral grounding devices” except as herein modified.

Resistor shall have the following characteristics:

- | | | |
|------|-----------------------------------|------------------------------|
| (1) | Impedance at 25°C | 5.0 ohms |
| (2) | Thermal current rating..... | 1200 rms symmetrical amperes |
| (3) | Mechanical current ratings..... | 1.2 times the thermal rating |
| (4) | Rated frequency | 50 Hz |
| (5) | Rated time | 10 seconds |
| (6) | Insulation class..... | “C” |
| (7) | Insulation class voltage | 15 kV |
| (8) | Temperature rise above 40°C | 500°C for steel |
| (9) | Dielectric test | Per ANSI/IEEE 32 or IEC |
| (10) | Service | Outdoor |

Remarks : Resistor ratings shall be verified by the Contractor and submitted to the Owner for approval.

The Contractor shall furnish and install a suitable copper bar between the low voltage neutral bushing and the resistor, and between the resistor and the ground pad on the cover.

4.6 Auxiliaries, Power and Controls

Two (2) feeds, each 400 volts, three-phase, 50 Hz, shall be provided for the auxiliary equipment's normal and alternate sources. The DC supplies for the tripping circuits shall be 220 volts DC nominal, with a voltage range of 188-246 volts.

An automatic transfer switch shall be provided to transfer the cooling equipment to the alternate source whenever the normal source fails.

5. PERFORMANCE TESTING AND GUARANTEES

All control panels and control cabinets shall be completely assembled and tested in the factory.

All tests shall be made in accordance with Technical Specification E-301 and the applicable IEC or ANSI standards.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for the unit auxiliary transformer shall conform to the requirements of Part 4 Section 2 Subsection 2.7.

The quality assurance/control procedures shall be as specified in Part 4 Section 2 Subsection 2.7. They shall include, but not limited to, the following:

- (1) Design criteria and engineering.
- (2) Material control.
- (3) Factory and site inspections.
- (4) Control and functional tests.
- (5) Wiring and insulation testing.
- (6) Insulation high voltage tests.
- (7) Protective relay and device settings and tests.
- (8) Welding.
- (9) Nondestructive testing.
- (10) Packaging and shipping.
- (11) Cleaning and painting procedure.

The quality assurance/quality control documentation as specified shall include, but not limited to, the following:

- (1) Inspection Reports.
- (2) Certified Test Reports.
- (3) Material Certifications.
- (4) All other documentation required by applicable codes and standards.

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